

LLMs Project

Chain-of-Thought Reasoning without Prompting

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Wang, X., & Zhou, D. (2023). *Chain-of-Thought Reasoning without Prompting*. Google DeepMind.
<https://arxiv.org/abs/2305.14325>

Introduction

- ❑ Prompting techniques often encode human priors, which makes it difficult to evaluate a model's *intrinsic* reasoning capabilities.
- ❑ Ideally, language models should be able to reason autonomously, without relying on handcrafted prompts or expensive fine-tuning.
- ❑ **Can large language models reason without prompting?**

CoT Decoding

Question in standard QA format

Q: I have 3 apples, my dad has 2 more apples than me, how many apples do we have in total?
A:

Language
model

Decoding step 0

top-1: 5
top-2: I
top-3: We
top-4: You
top-5: The

Continue greedy decoding

5 apples ✗

I have 3 apples, my dad has 2 more apples than me, so he has 5 apples. $3+5=8$. We have 8 apples in total. ✓

We have 5 apples in total. ✗

You have 3 apples, your dad has 2 more apples than you, so he has 5 apples. $3+5=8$. You have 8 apples in total. ✓

The answer is 5. ✗

uncertain

certain

Illustration of CoT-decoding. Pre-trained LLMs are capable of inherent reasoning without prompting by considering alternative top- k tokens, rather than solely relying on the top-1 greedy decoding path. Moreover, these models tend to display higher confidence in decoding the final answer (indicated by a darker shaded color) when a CoT reasoning path is present.

Extracting CoT Paths

Problem:

CoT paths are often not the most probable → standard ranking or self-consistency fails without prompts.

Key Idea:

CoT paths yield more confident answers. Confidence is measured using **token-level probability margins**.

$$\Delta_{k,\text{answer}} = \frac{1}{|\text{answer}|} \sum_{x_t \in \text{answer}} p(x_t^1 | x_{<t}) - p(x_t^2 | x_{<t}).$$

Path Confidence Margin Calculation Example

- ▶ **Decoding Output:** "64" (2 tokens: "6" and "4")
- ▶ **For Token "6":**
 - ▶ $p('6') = 0.91$
 - ▶ $p(2\text{nd best}) = 0.05$
 - ▶ **Margin:** $0.91 - 0.05 = 0.86$
- ▶ **For Token "4":**
 - ▶ $p('4') = 0.88$
 - ▶ $p(2\text{nd best}) = 0.10$
 - ▶ **Margin:** $0.88 - 0.10 = 0.78$
- ▶ **Overall Margin Calculation:**

$$\Delta_{k,\text{answer}} = \frac{1}{2} (0.86 + 0.78) = 0.82$$

Extracting CoT Paths

Definitions:

Definition 1 (Token Sequence). Let $\mathcal{V}$ be the vocabulary of tokens. A sequence $X = (x_1, x_2, \dots, x_n)$ with $x_i \in \mathcal{V}$ represents a token sequence.

Definition 2 (Language Model). A language model $\mathcal{M}$ is a function that maps a token sequence to a probability distribution over the next token: $\mathcal{M} : \mathcal{V}^* \rightarrow \Delta(\mathcal{V})$, where $\Delta(\mathcal{V})$ is the probability simplex over $\mathcal{V}$.

Definition 3 (Sequence Probability). For a token sequence $X = (x_1, x_2, \dots, x_n)$, we define the probability of generation under model $\mathcal{M}$ as:

$$P_{\mathcal{M}}(X) = \prod_{i=1}^n P_{\mathcal{M}}(x_i | x_1, \dots, x_{i-1}) \quad (1)$$

Definition 4 (Problem Components). Given a reasoning problem Q , let R be a reasoning path (sequence of tokens representing intermediate reasoning steps) and A be the answer tokens. The complete generation is $X = Q \oplus R \oplus A$, where $\oplus$ denotes sequence concatenation.

Extracting CoT Paths

Definitions:

Definition 5 (Token Confidence). *The confidence score for token x_t is defined as:*

$$\Delta(x_t|x_{<t}) = P_{\mathcal{M}}(x_t^1|x_{<t}) - P_{\mathcal{M}}(x_t^2|x_{<t}) \quad (2)$$

Definition 6 (Answer Confidence). *The overall confidence score for an answer $A = (a_1, a_2, \dots, a_m)$ is:*

$$\Delta_A = \frac{1}{m} \sum_{i=1}^m \Delta(a_i|x_{<a_i}) \quad (3)$$

Definition 7 (Next-Token Entropy). *The entropy of the next-token distribution at position t is:*

$$H_t = - \sum_{v \in \mathcal{V}} P_{\mathcal{M}}(v|x_{<t}) \log P_{\mathcal{M}}(v|x_{<t}) \quad (4)$$

Extracting CoT Paths

Confidence-Entropy Relationship:

The confidence score $\Delta(x_t|x_{<t})$ is inversely related to the entropy H_t .

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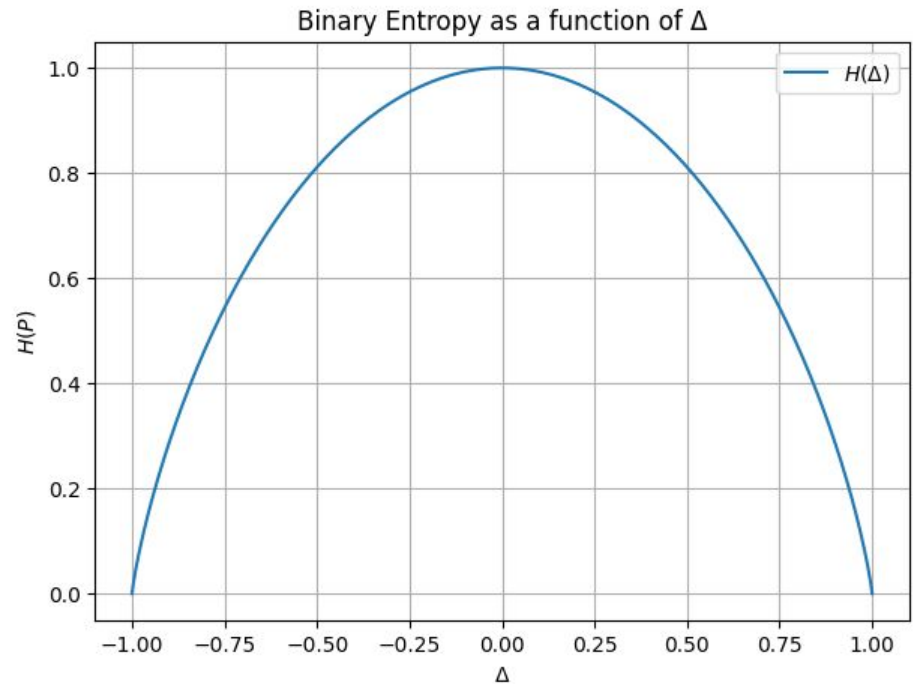
$$H(P) = -\frac{1 + \Delta}{2} \log \frac{1 + \Delta}{2} - \frac{1 - \Delta}{2} \log \frac{1 - \Delta}{2}.$$

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Extracting CoT Paths

Confidence-Entropy Relationship:

mutual information

$$I(R; A|Q) = H(A|Q) - H(A|Q, R)$$

Extracting CoT Paths

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mutual information $I(R; A|Q) = H(A|Q) - H(A|Q, R)$

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$$I(R; A|Q) > 0$$

$$H(A|Q, R) < H(A|Q)$$

$$\mathbb{E}[\Delta_A|Q, R] > \mathbb{E}[\Delta_A|Q]$$

Experiments

We try to reproduce the result and show that COT decoding can improve accuracy of the model on reasoning task.

But also using variant : What if we use **beams** search, **top p** or **top k** decoding instead of greedy ?

We use the dataset ChilleD/MultiArith from Huggingface.
example :

Q : There are 64 students trying out for the school's trivia teams. If 36 of them didn't get picked for the team and the rest were put into 4 groups, how many students would be in each group?

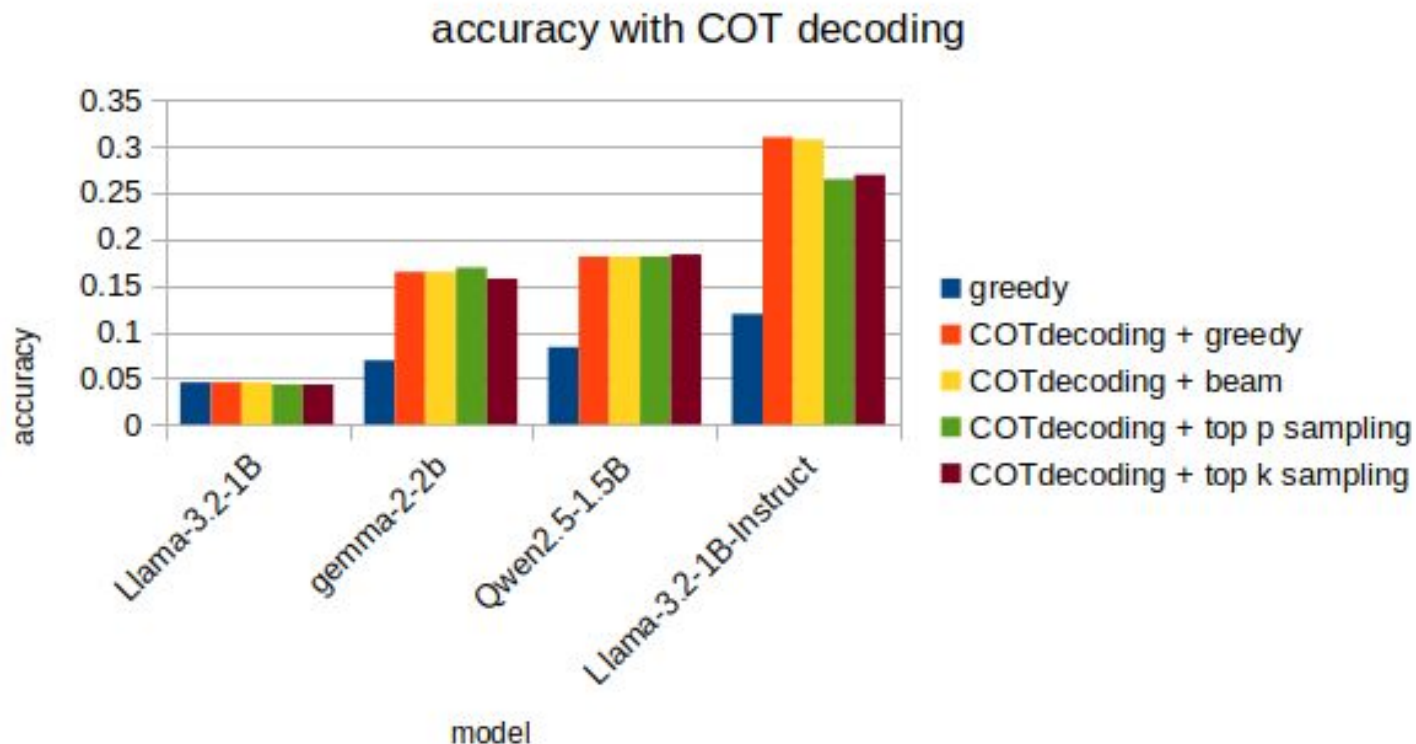
A : 7

Experiments

Greedy vs CoT-decoding variants.

Gemma-2, Llama3.2 and Qwen2.5, Llama3.2 instruct

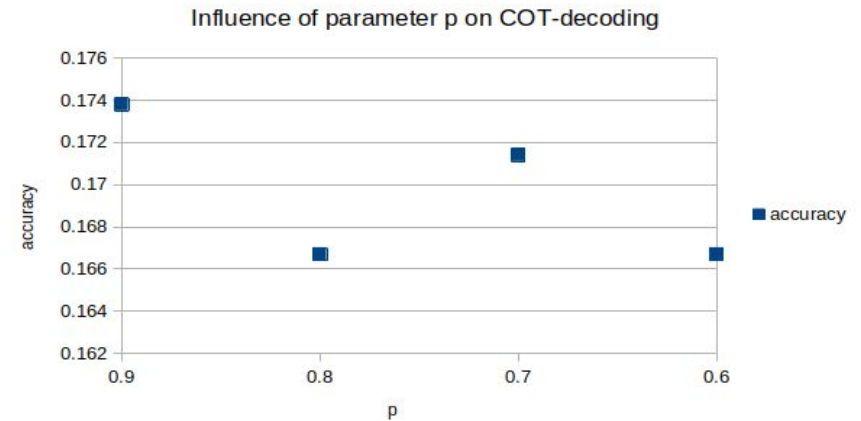
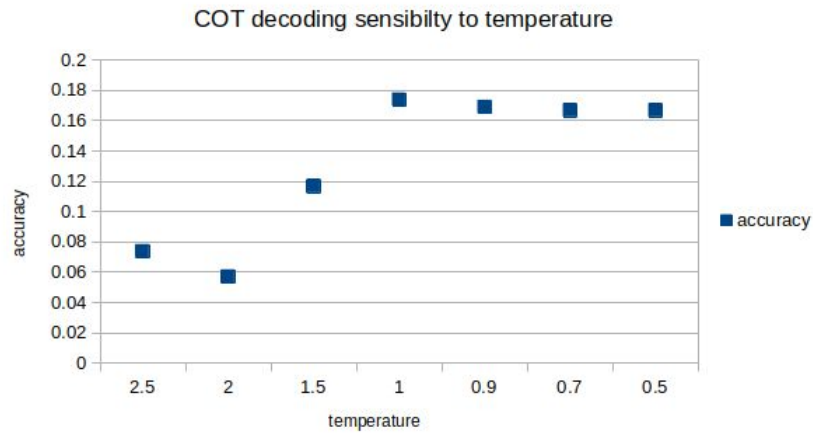
Accuracy of the LLM with different COT decoding.



Experiments

Influence of temperature in the sampling variant ?

Influence of p in the sampling variant ?



Score Evolution : Motivation

CoT decoding typically computes a score only for the **final answer span**, not for intermediate reasoning steps.

How does the score evolve over time as tokens are generated during reasoning?

Can the trajectory of token-level scores indicate something meaningful?

- Can we identify and **skip** low-quality reasoning branches **early**?
- Can we cluster score evolution **patterns** to distinguish correct from incorrect answers?

Objective: Explore score evolution as a signal to improve CoT decoding efficiency and accuracy.

Score Evolution : Method

Compute the **per-token margin**:

$$\delta_t = p(x_t^1 \mid x_{<t}) - p(x_t^2 \mid x_{<t})$$

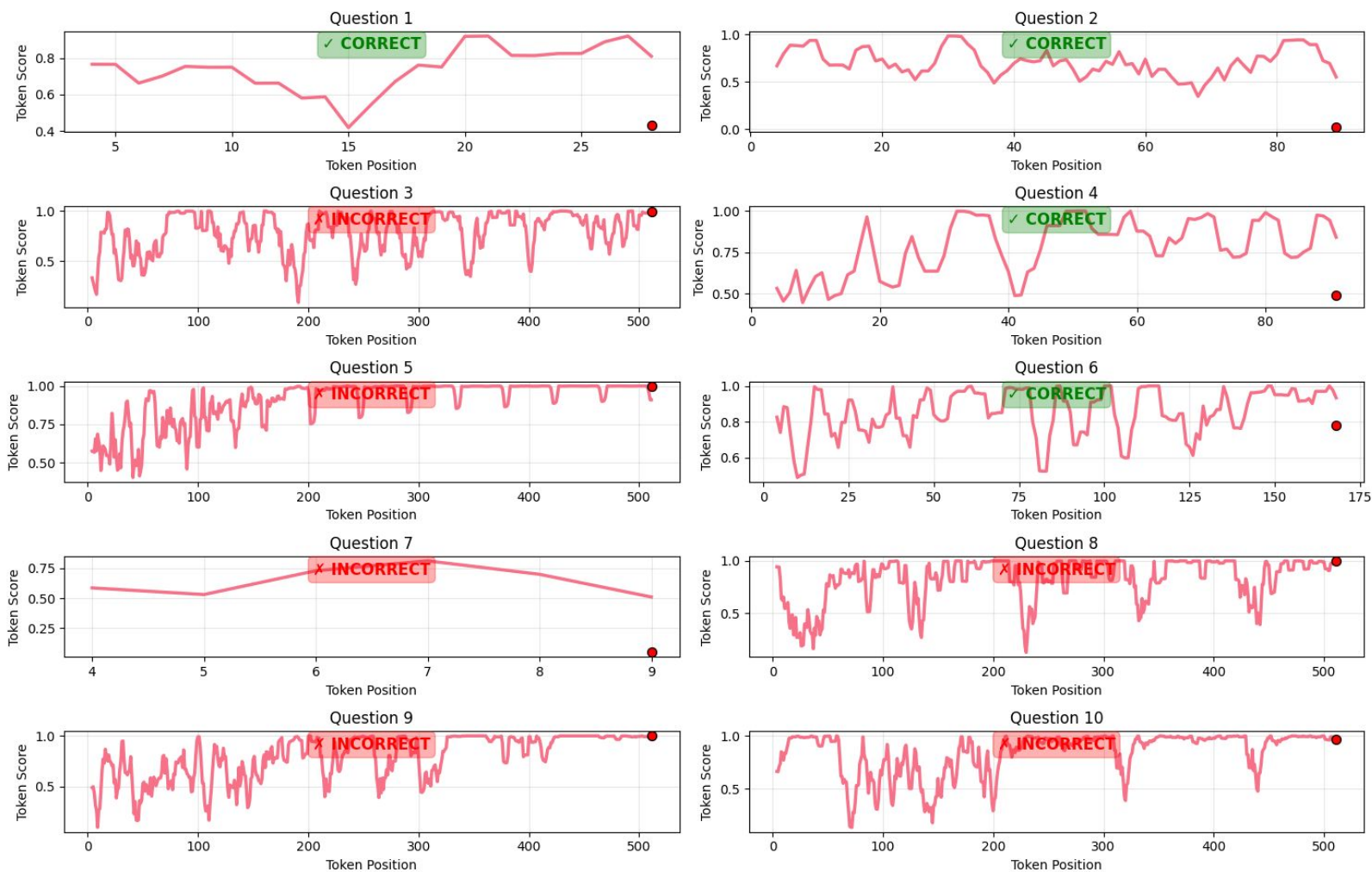
Plot δ_t on the y-axis and token index t (within the answer) on the x-axis.

Score Evolution : Pipeline

- Load first 10 questions from GSM8K
- Generate top-3 CoT reasoning paths
- Compute token-level scores
- Check correctness by comparing final numbers
- Plot score evolution with moving average
- Highlight final score and correctness

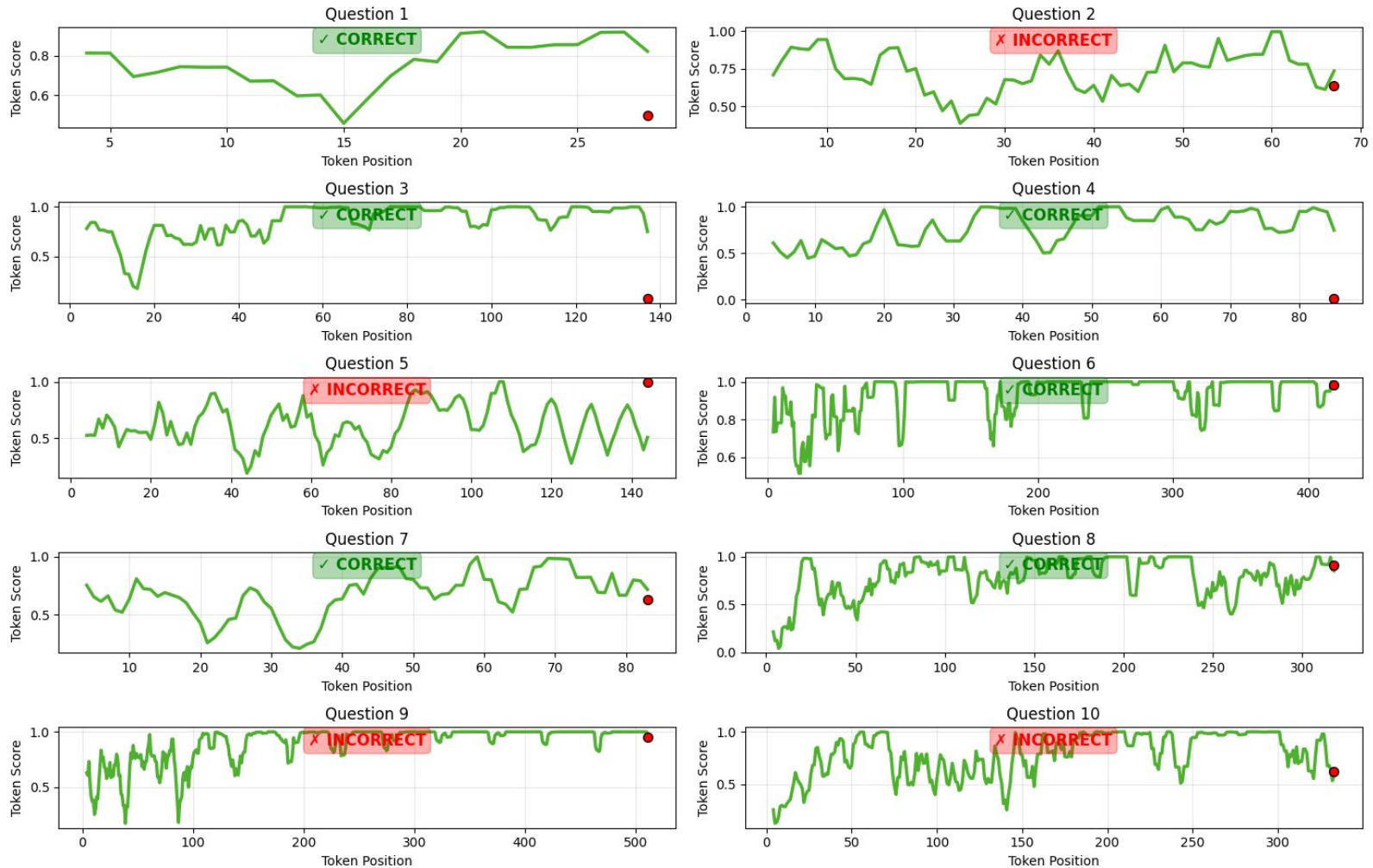
Score Evolution: Results

Score Evolution for $k=1$ Across All Questions



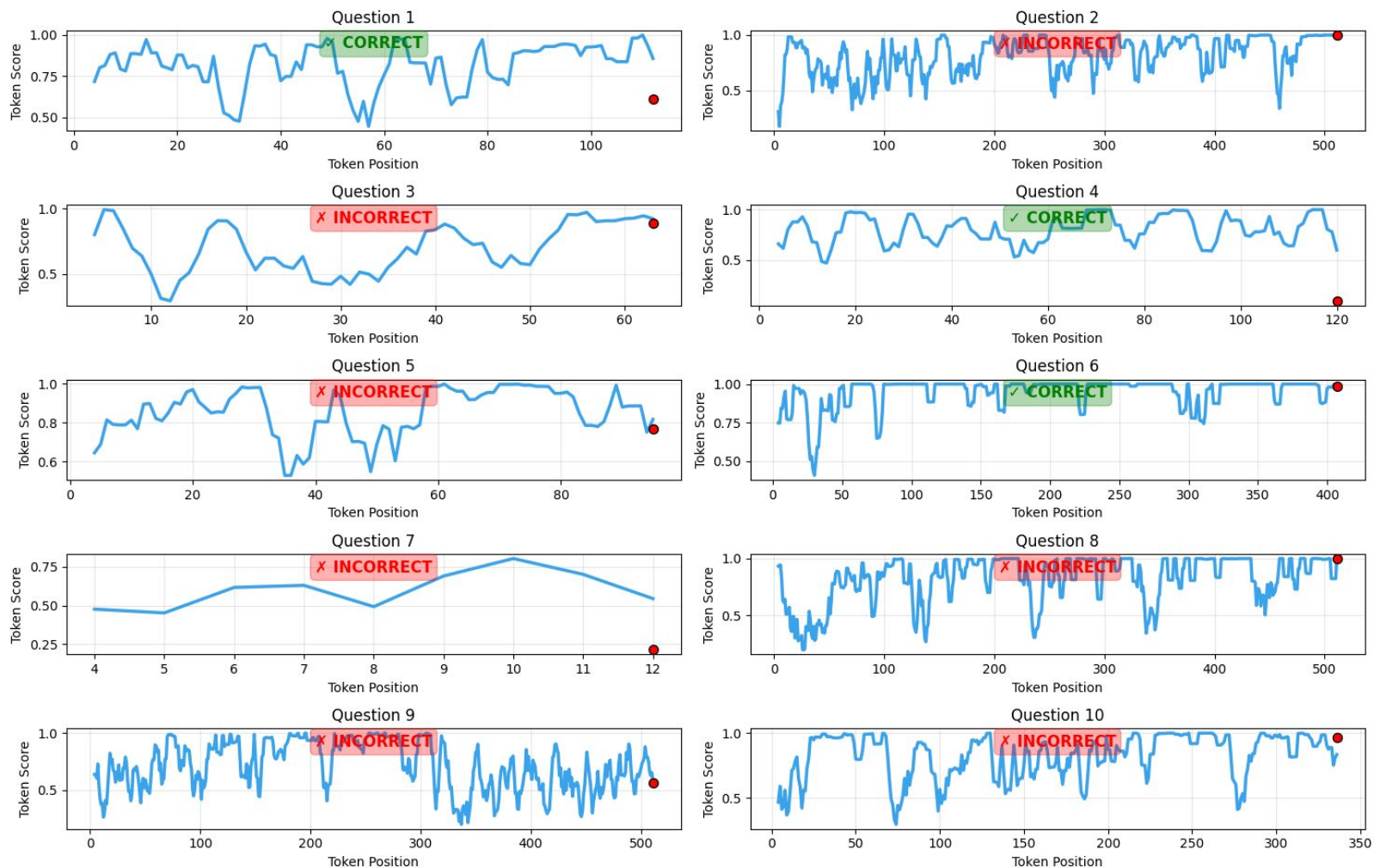
Score Evolution: Results

Score Evolution for k=2 Across All Questions



Score Evolution: Results

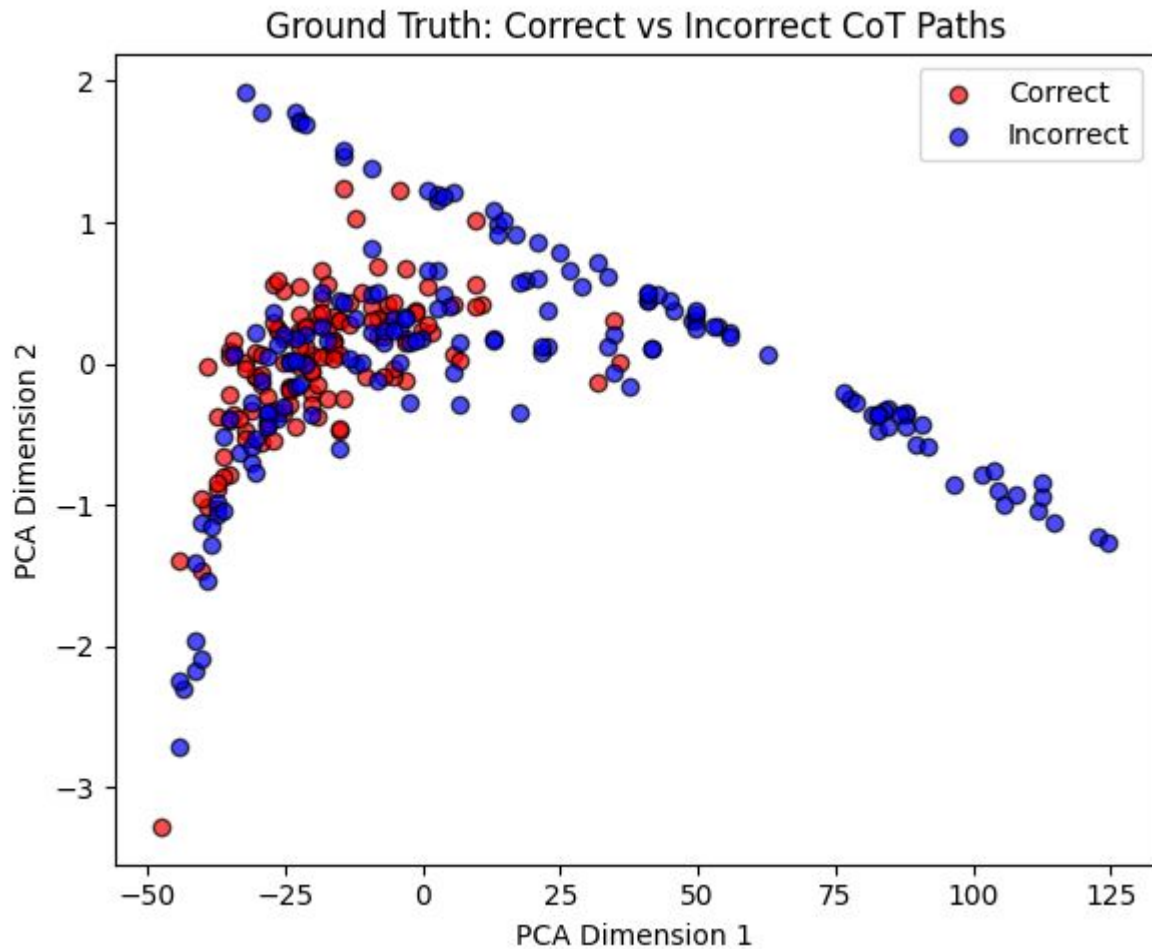
Score Evolution for $k=3$ Across All Questions



Score Evolution: Clustering

- Use 100 GSM8K questions
- Extract 9 features from token score sequences (e.g. slope, entropy..)
- Apply **KMeans** / **GMM (2 clusters)** to group correct vs incorrect paths
- Evaluate with accuracy against ground-truth labels
- Use **PCA** for 2D visualization of clusters

Score Evolution: Clustering



We can Filter out some likely incorrect paths

Conclusion

In our project we :

- Reproduced CoT decoding results from the original paper
 - Implementation
 - Show gains on based and instruct model
- Tested some CoT decoding variants not included in the paper (beams and samplings)
- Proposed a mathematical intuition
- Studied **token-level score evolution** as a signal for answer correctness